Problem Set 3

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Contents

Question 1 **2**

Question 2 **5**

Question 3 **7**

Question 4 **9**

Question 5 **11**

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

(1) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

Answer: First, we import the data and check the three variables for missing data.

```
library(stargazer)
incumbents <- read.csv("incumbents_subset.csv", row.names = 1)
incumbents <- incumbents[c("voteshare", "difflog", "presvote")]
descriptive <- rbind(Elections = nrow(incumbents),
                     Missing = colSums(is.na(incumbents)),
                     sapply(incumbents, summary))
stargazer(t(descriptive),  # Create the descriptive statistics table.
          summary = FALSE, digits = 4, float = FALSE)
```

	Elections	Missing	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
voteshare	3,193	0	0.3476	0.5846	0.6569	0.6552	0.7233	0.9930
difflog	3,193	0	-3.0601	0.6734	1.6106	1.8278	2.9857	5.8558
presvote	3,193	0	0.1931	0.4695	0.5478	0.5512	0.6240	0.9606

All 3,193 elections have complete data (no NAs).

We regress incumbent `voteshare` $\sim$ `difflog` (log incumbent spending minus log challenger spending). The regression tables give standard errors in parentheses.

```
vote_spending <- lm(voteshare ~ difflog, data = incumbents)
stargazer(vote_spending, digits = 6, single.row = TRUE,  # Create regression table.
          report = "vc*st", omit.stat = c("ser", "f"), float = FALSE)
```

	Dependent variable:
	voteshare
difflog	0.041666*** (0.000968) t = 43.044060
Constant	0.579031*** (0.002251) t = 257.188300
Observations	3,193
R^2	0.367341
Adjusted R^2	0.367143
Note:	*p<0.1; **p<0.05; ***p<0.01

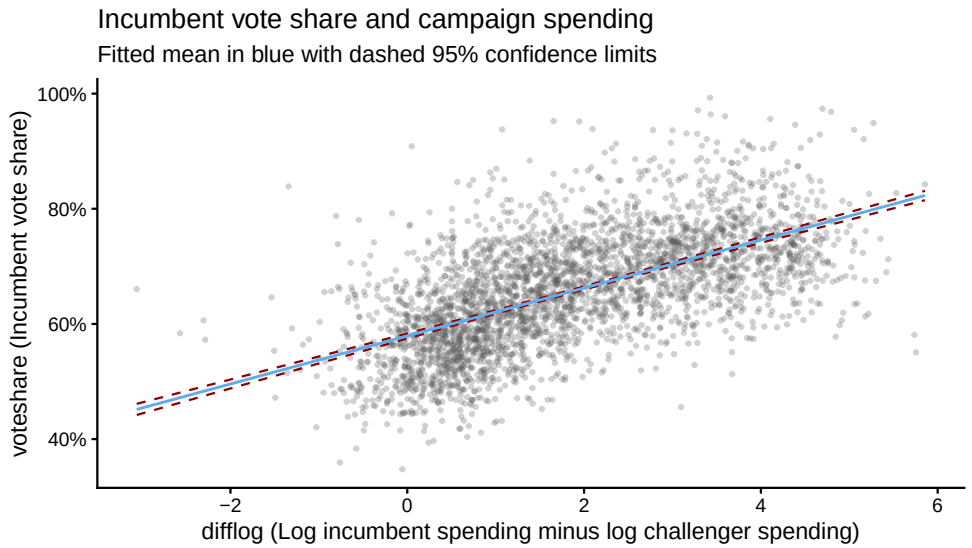
The spending gap explains 36.73% of the variation in incumbent vote share. The slope is positive. Our null hypothesis is that the population spending-gap slope is zero. Our alternative is that it differs from zero. The slope is positive ($t = 43.04$, two-sided $p < 0.001$), so we reject the null at 5%.

(2) Make a scatterplot of the two variables and add the regression line.

Answer: First, we calculate the fitted mean and its pointwise 95% confidence limits across the observed spending gaps (to use later in the scatterplot).

```
library(ggplot2)
spending_gap <- seq(min(incumbents$difflog), max(incumbents$difflog),
                    length.out = 200)
vote_grid <- data.frame(difflog = spending_gap)
vote_grid <- cbind(vote_grid, predict(vote_spending, vote_grid,
                              interval = "confidence", level = 0.95))
```

```
vote_spending_plot <- ggplot(vote_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "difflog (Log incumbent spending minus log challenger spending)",
       y = "voteshare (Incumbent vote share)") +
  theme_classic(base_size = 12)
ggsave("PS03_Q1_vote_spending.pdf", vote_spending_plot,
       width = 7, height = 4)
```



Incumbents tend to receive a larger vote share as their spending advantage over the challenger grows. The spending gap explains 36.73% of the variation in incumbent vote share. Individual election results vary much more widely around the fitted line than the dashed 95% confidence limits for the mean.

(3) Save the residuals of the model in a separate object.

Answer: We save the residuals and check the first election. A residual is the observed vote share minus the fitted vote share.

```
vote_resid <- resid(vote_spending)
first_election <- c(observed = incumbents$voteshare[1],
                    fitted = unname(fitted(vote_spending)[1]),
                    residual = unname(vote_resid[1]))
stargazer(t(first_election), summary = FALSE, digits = 6, float = FALSE)  # Create output table.
```

observed	fitted	residual
0.602361	0.602784	-0.000423

The first incumbent received 60.24% of the vote, about 0.042 percentage points below the prediction. This explains the negative residual.

(4) Write the prediction equation.

Answer: We take the intercept and slope from the fitted model:

```
equation <- round(coef(vote_spending), 6) # Rounded to 6 decimals.
stargazer(t(equation), summary = FALSE, digits = 6, float = FALSE)  # Create coefficient table.
```

(Intercept)	difflog
0.579031	0.041666

For election i, the prediction equation is:

$$\widehat{\texttt{voteshare}}_i = 0.579031 + 0.041666\,\texttt{difflog}_i$$

On average, a one-unit change in the log spending gap in the incumbent's favour predicts a gain of 4.17 percentage points in incumbent vote share, since the fitted slope is $\hat{\beta}_1 = 0.041666$. At equal spending ($\texttt{difflog} = 0$), the intercept $\hat{\beta}_0 = 0.579031$ gives a predicted incumbent vote share of 57.90%.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

(1) Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

Answer: We will use `difflog` to predict the vote share of the presidential candidate from the incumbent's party (`presvote`).

```
pres_spending <- lm(presvote ~ difflog, data = incumbents)
stargazer(pres_spending, digits = 6, single.row = TRUE,  # Create regression table.
          report = "vc*st", omit.stat = c("ser", "f"), float = FALSE)
```

	Dependent variable:
	presvote
difflog	0.023837*** (0.001359)
	t = 17.541820
Constant	0.507583*** (0.003161)
	t = 160.600800
Observations	3,193
R^2	0.087951
Adjusted R^2	0.087665
Note:	*p<0.1; **p<0.05; ***p<0.01

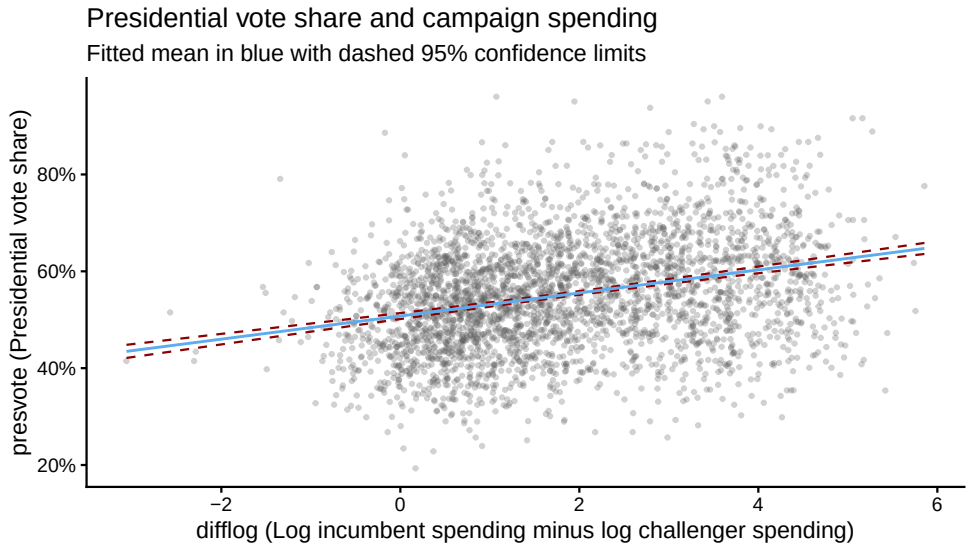
Our null hypothesis is that the population spending-gap slope is zero. Our two-sided alternative is that it differs from zero. We reject the null at 5% ($t = 17.54$, $p < 0.001$). The association is positive. The gap explains 8.80% of the variation in presidential vote share.

(2) Make a scatterplot of the two variables and add the regression line.

Answer: We put `difflog` on the horizontal axis and `presvote` on the vertical axis. We calculate the 95% confidence limits from the model in part (1).

```
pres_grid <- data.frame(difflog = spending_gap)
pres_grid <- cbind(pres_grid, predict(pres_spending, pres_grid,
                           interval = "confidence", level = 0.95))
```

```
pres_spending_plot <- ggplot(pres_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = presvote),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Presidential vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "difflog (Log incumbent spending minus log challenger spending)",
       y = "presvote (Presidential vote share)") +
  theme_classic(base_size = 12)
ggsave("PS03_Q2_pres_spending.pdf", pres_spending_plot,
       width = 7, height = 4)
```



The line slopes upward, but the points are more spread out than in Question 1. The spending gap tells us less about the presidential vote.

(3) Save the residuals of the model in a separate object.

Answer: We save the presidential residuals and check the first election:

```
pres_resid <- resid(pres_spending)
first_election <- c(observed = incumbents$presvote[1],
                    fitted = unname(fitted(pres_spending)[1]),
                    residual = unname(pres_resid[1]))
stargazer(t(first_election), summary = FALSE, digits = 6, float = FALSE)  # Create output table.
```

observed	fitted	residual
0.526778	0.521173	0.005606

The first residual is positive: the presidential candidate received 0.56 percentage points more than predicted.

(4) Write the prediction equation.

Answer: We can read the two coefficients from the model:

```
equation <- round(coef(pres_spending), 6) # Rounded to 6 decimals.
stargazer(t(equation), summary = FALSE, digits = 6, float = FALSE)  # Create coefficient table.
```

(Intercept)	difflog
0.507583	0.023837

The prediction equation is

$$\widehat{\texttt{presvote}}_i = 0.507583 + 0.023837\,\texttt{difflog}_i$$

On average, a one-unit change in `difflog` from 0 to 1 predicts a 2.38-percentage-point rise in the party's presidential vote share, since the slope is $\hat{\beta}_1 = 0.023837$. At equal spending, the intercept $\hat{\beta}_0 = 0.507583$ gives a predicted presidential vote share of 50.76%.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

(1) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

Answer: We regress `voteshare` ~ `presvote`:

```
vote_pres <- lm(voteshare ~ presvote, data = incumbents)
stargazer(vote_pres, digits = 6, single.row = TRUE,  # Create regression table.
          report = "vc*st", omit.stat = c("ser", "f"), float = FALSE)
```

	Dependent variable:
	voteshare
presvote	0.388018*** (0.013493)
	t = 28.756740
Constant	0.441330*** (0.007599)
	t = 58.080330
Observations	3,193
R^2	0.205814
Adjusted R^2	0.205565
Note:	*p<0.1; **p<0.05; ***p<0.01

Our null hypothesis is that the population slope on presidential vote share is zero. Our two-sided alternative is that it differs from zero. We reject the null at the 5% level ($t = 28.76$, $p < 0.001$). Greater support for the party's presidential candidate predicts greater support for its incumbent. Presidential vote share explains 20.58% of the variation in incumbent vote share.

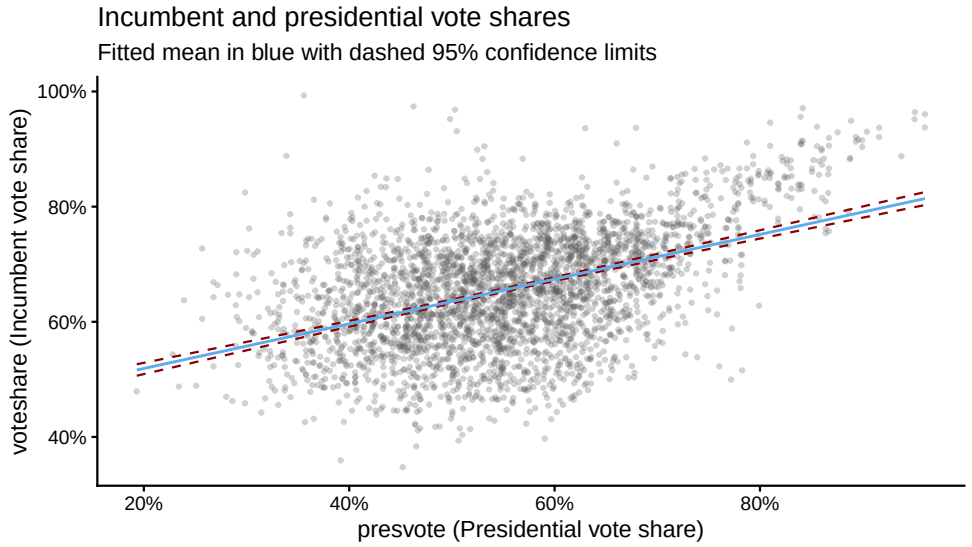
(2) Make a scatterplot of the two variables and add the regression line.

Answer: We plot one point per election and show both vote shares as percentages. The presidential vote share nearest 50% in the grid for the 95% confidence limits is 50.17%.

```
pres_share <- seq(min(incumbents$presvote), max(incumbents$presvote),
                  length.out = 200)
support_grid <- data.frame(presvote = pres_share)
support_grid <- cbind(support_grid, predict(vote_pres, support_grid,
                                  interval = "confidence", level = 0.95))
```

```
vote_pres_plot <- ggplot(support_grid, aes(x = presvote)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_x_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent and presidential vote shares",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "presvote (Presidential vote share)",
       y = "voteshare (Incumbent vote share)") +
  theme_classic(base_size = 12)
```

```
ggsave("PS03_Q3_vote_pres.pdf", vote_pres_plot,
       width = 7, height = 4)
```



Incumbents tend to receive a larger share of the vote where their party's presidential candidate does better. Even near a presidential vote share of 50%, we see incumbents who lost their race as well as incumbents who won comfortably.

(3) Write the prediction equation.

Answer: We use the two coefficients from the fitted line:

```
equation <- round(coef(vote_pres), 6) # Rounded to 6 decimals.
stargazer(t(equation), summary = FALSE, digits = 6, float = FALSE)  # Create coefficient table.
```

(Intercept)	presvote
0.441330	0.388018

The prediction equation is

$$\widehat{\texttt{voteshare}}_i = 0.441330 + 0.388018\,\texttt{presvote}_i$$

On average, a one-unit change in presidential vote share (100 percentage points) predicts a 38.80-percentage-point change in incumbent vote share in the same direction, since the slope is $\hat{\beta}_1 = 0.388018$. A rise from 50% to 60% therefore predicts a gain of $0.388018(0.60 - 0.50) \approx 0.03880$, or 3.88 percentage points, for the incumbent. The intercept $\hat{\beta}_0 = 0.441330$ predicts 44.13% incumbent vote share at zero presidential support. That baseline is an extrapolation because the smallest presidential vote share in the sample is 19.31%.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

(1) Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

Answer: We can use the incumbent vote-share residuals saved in Question 1(3) as the outcome and the presidential vote-share residuals saved in Question 2(3) as the explanatory variable.

```
residual_data <- data.frame(vote_resid, pres_resid)
resid_model <- lm(vote_resid ~ pres_resid, data = residual_data)
stargazer(resid_model, digits = 6, single.row = TRUE,  # Create regression table.
          report = "vc*st", omit.stat = c("ser", "f"), float = FALSE)
```

	Dependent variable:
	vote_resid
pres_resid	0.256877*** (0.011762)
	t = 21.839750
Constant	−0.000000 (0.001299)
	t = −0.000000
Observations	3,193
R^2	0.130038
Adjusted R^2	0.129765
Note:	*p<0.1; **p<0.05; ***p<0.01

Our null hypothesis is that the population slope relating the two spending-adjusted shares is zero. Our two-sided alternative is that it differs from zero. We reject the null at the 5% level ($t = 21.84$, $p < 0.001$). The presidential residual explains 13.00% of the variation left in incumbent vote share after adjusting for spending.

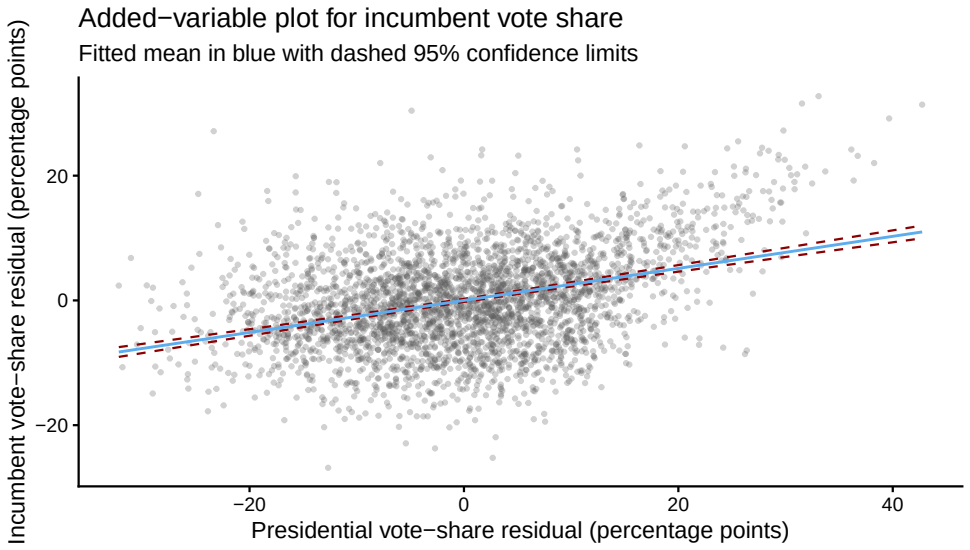
(2) Make a scatterplot of the two residuals and add the regression line.

Answer: We put the presidential residual on the horizontal axis and the incumbent residual on the vertical axis.

```
pres_gap <- seq(min(residual_data$pres_resid), max(residual_data$pres_resid),
                length.out = 200)
resid_grid <- data.frame(pres_resid = pres_gap)
resid_grid <- cbind(resid_grid, predict(resid_model, resid_grid,
                                        interval = "confidence", level = 0.95))
```

```
resid_plot <- ggplot(resid_grid, aes(x = pres_resid)) +
  geom_point(data = residual_data, aes(y = vote_resid),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  # To read the residuals in percentage points, we multiply the axis labels by 100.
```

```
  scale_x_continuous(labels = scales::label_number(scale = 100)) +
  scale_y_continuous(labels = scales::label_number(scale = 100)) +
  labs(title = "Added−variable plot for incumbent vote share",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Presidential vote−share residual (percentage points)",
       y = "Incumbent vote−share residual (percentage points)") +
  theme_classic(base_size = 12)
ggsave("PS03_Q4_added_variable.pdf", resid_plot,
       width = 7, height = 4)
```



Each point represents one election. The fitted line slopes upward. After adjusting both vote shares for spending, incumbents tend to do better than predicted where their party's presidential candidate also does better than predicted. The first election lies in the lower-right quadrant, with presidential vote share 0.56 percentage points above its prediction and incumbent vote share 0.042 percentage points below its prediction.

(3) Write the prediction equation.

Answer: We take the coefficients from the residual regression:

```
equation <− round(coef(resid_model), 6) # Rounded to 6 decimals.
stargazer(t(equation), summary = FALSE, digits = 6, float = FALSE)  # Create coefficient table.
```

(Intercept)	pres_resid
0	0.256877

$$\widehat{\texttt{vote_resid}}_i = 0.256877\,\texttt{pres_resid}_i$$

On average, a one-unit change in the presidential vote-share residual predicts a 0.256877-unit change in the incumbent vote-share residual in the same direction, as measured by the slope $\hat{\beta}_1$. In percentage points, a presidential residual one point higher predicts an incumbent residual about 0.257 points higher. Both sets of residuals average zero because their original regressions include intercepts. The fitted intercept $\hat{\beta}_0$ therefore rounds to zero, and the line passes through the origin.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

(1) Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

Answer: We will include the spending gap and presidential vote share in the same regression. Each slope measures the association with incumbent vote share while we hold the other predictor fixed.

```
full_model <- lm(voteshare ~ difflog + presvote, data = incumbents)
stargazer(full_model, digits = 6, single.row = TRUE,  # Create regression table.
          report = "vc*st", omit.stat = c("ser", "f"), float = FALSE)
```

	Dependent variable:
	voteshare
difflog	0.035543*** (0.000946)
	t = 37.590140
presvote	0.256877*** (0.011764)
	t = 21.836330
Constant	0.448644*** (0.006330)
	t = 70.879480
Observations	3,193
R^2	0.449610
Adjusted R^2	0.449265
Note:	*p<0.1; **p<0.05; ***p<0.01

For each predictor, our null hypothesis is that its population slope is zero with the other predictor held fixed. The two-sided alternative is that its slope differs from zero. We reject both null hypotheses at the 5% level ($p < 0.001$ for each slope). Together the predictors explain 44.96% of the variation in incumbent vote share.

(2) Write the prediction equation.

Answer: We use the fitted intercept and both partial slopes:

```
equation <- round(coef(full_model), 6) # Rounded to 6 decimals.
stargazer(t(equation), summary = FALSE, digits = 6, float = FALSE)  # Create coefficient table.
```

(Intercept)	difflog	presvote
0.448644	0.035543	0.256877

Therefore, the prediction equation is

$$\widehat{\texttt{voteshare}}_i = 0.448644 + 0.035543\,\texttt{difflog}_i + 0.256877\,\texttt{presvote}_i$$

At equal spending and zero presidential vote share, the intercept $\hat{\beta}_0 = 0.448644$ gives a predicted incumbent vote share of 44.86%. On average, a one-unit change in the log spending gap in the incumbent's favour predicts a gain of 3.55 percentage points in incumbent vote share ($\hat{\beta}_1 = 0.035543$), holding presidential vote share constant. Holding the log spending gap constant, the slope $\hat{\beta}_2 = 0.256877$ gives the predicted gain in incumbent vote share for a one-unit rise in presidential vote share. A rise from 50% to 60% therefore predicts a 2.57-percentage-point gain for the incumbent, compared with 3.88 points in Question 3.

(3) What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

Answer: The coefficient on `presvote` in Question 5 equals the coefficient on `pres_resid` in Question 4 to numerical precision. The left column shows the residual regression from Question 4 and the right column shows the full regression from Question 5.

```
old_options <- options(digits = 15)  # Show extra digits for the two slopes.
stargazer(resid_model, full_model,  # Create the model comparison table.
          digits = 10, single.row = TRUE,
          omit.stat = "f", float = FALSE)
```

	Dependent variable:	
	vote_resid	voteshare
	(1)	(2)
pres_resid	0.2568770127*** (0.0117619024)	
difflog		0.0355430864*** (0.0009455428)
presvote		0.2568770127*** (0.0117637458)
Constant	−0.0000000000 (0.0012986050)	0.4486442218*** (0.0063296774)
Observations	3,193	3,193
R^2	0.1300376195	0.4496104332
Adjusted R^2	0.1297649895	0.4492653614
Residual Std. Error	0.0733797985 (df = 3191)	0.0733912991 (df = 3190)
Note:		*p<0.1; **p<0.05; ***p<0.01

We compare the unrounded presidential slopes first, then the residuals for all 3,193 elections.

```
print(all.equal(coef(resid_model)[["pres_resid"]], # Comparing slopes.
                coef(full_model)[["presvote"]]))
print(all.equal(resid(resid_model), resid(full_model))) # Comparing residuals.
options(old_options)
```

```
[1] TRUE
[1] TRUE
```

Both slopes equal 0.2568770127, and the residuals match to numerical precision. For a one-percentage-point increase in presidential vote share at the same spending gap, the model predicts about 0.257 percentage points more incumbent vote share.

For the explanation of why this happens, we can see the slides 12–14 of the Week 9 tutorial: In Question 1, we subtract the incumbent vote share predicted by spending from the observed incumbent share. In Question 2, we do the same for the presidential share, using the same spending gap and the same elections. What remains in each share is uncorrelated with the spending gap. Regressing the first residual on the second therefore uses exactly the variation that estimates the presidential slope when we hold spending fixed in Question 5. This is the Frisch–Waugh–Lovell result.

The standard errors both round to 0.01176. For the error variance, R divides the same residual sum of squares by 3,191 in Question 4 and by 3,190 in Question 5, where we estimate one more coefficient. This gives the slightly larger standard error in Question 5. The R^2 values use different totals as well. Question 4 accounts for 13.00% of the incumbent variation remaining after spending, and Question 5 accounts for 44.96% of the original variation.